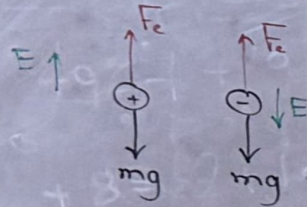
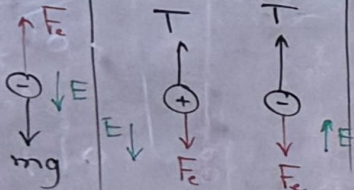


بسم الله الرحمن الرحيم

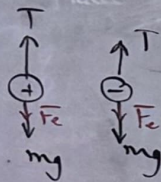
① $T = 0$



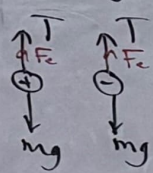
② $mg = 0$



$T \neq 0, T > mg$



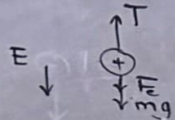
$T \neq 0, mg > T$



$$\begin{aligned}
 m, g &\rightarrow 10^{-3} \\
 \mu &\rightarrow 10^{-6} \\
 n &\rightarrow 10^9 \\
 \text{cm} &\div 100 \\
 \text{cm}^2 &= \times 10^{-4}
 \end{aligned}$$

ex. 0.6 g charge is placed in $E = 300 \text{ N/C}$ downward.

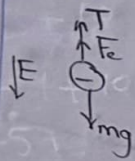
(a) Tension when $q = 8 \mu\text{C}$



$$T = mg + qE$$

$$= 0.6 \times 10^{-3} \times 9.8 + 8 \times 10^{-6} \times 300 = 0.00828 = \underline{8.3 \text{ mN}}$$

(b) Tension when $q = -8 \mu\text{C}$



$$T + qE = mg$$

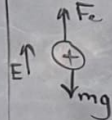
$$T = mg - qE$$

$$= 0.6 \times 10^{-3} \times 9.8 - 8 \times 10^{-6} \times 300 = \underline{3.8 \text{ mN}}$$

Q14

0.2 g charge is placed in 2000 N/C upward

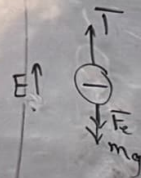
The charge when $T = 0$:



$$qE = mg$$

$$q = \frac{mg}{E} = \frac{0.2 \times 10^{-3} \times 9.8}{2000} = \underline{0.65 \mu\text{C}}$$

The charge when $T = 4 \text{ mN}$



$$T = mg + qE$$

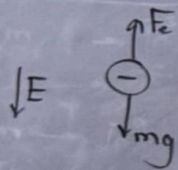
$$T - mg = qE$$

$$\frac{T - mg}{E} = q \Rightarrow \underline{0.68 \mu\text{C}} \rightarrow \underline{-0.68 \mu\text{C}}$$

$$\begin{aligned}
 mg &= 0.2 \times 10^{-3} \times 9.8 \\
 &= 1.96 \times 10^{-3} = \underline{1.96 \text{ mN}}
 \end{aligned}$$

Q₉ 0.2 kg charge is placed in $E = 2 \text{ N/C}$ downward

The charge is:



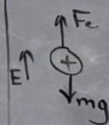
$$qE = mg$$

$$q = \frac{mg}{E} = \frac{0.2 \times 10}{2} = 1 \text{ C} \rightarrow q = -1 \text{ C}$$

Q₁₄

0.2 g charge is placed in 2000 N/C upward

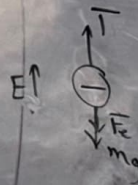
The charge when $T = 0$:



$$qE = mg$$

$$q = \frac{mg}{E} = \frac{0.2 \times 10^{-3} \times 9.8}{2000} = 0.65 \mu\text{C}$$

The charge when $T = 4 \text{ mN}$



$$mg = 0.2 \times 10^{-3} \times 9.8 = 1.96 \times 10^{-3} = 1.96 \text{ mN}$$

$$T = mg + qE$$

$$T - mg = qE$$

$$\frac{T - mg}{E} = q \Rightarrow 0.68 \mu\text{C} \rightarrow -0.68 \mu\text{C}$$

بسم الله الرحمن الرحيم

float: $\rho_o h_o = \rho_f h_f$

$h_f \rightarrow$ depth

$h_o =$

$h_d =$ over - above

$$V_o = \frac{\omega_o}{\rho_o g}$$

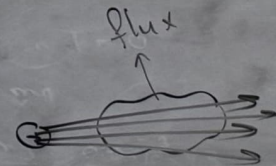
full immersed

$$F_b = \frac{\omega_o \rho_f}{\rho_o}$$

$$V_o = \frac{\omega_o}{\rho_o g}$$

$$F_b = \rho_f V_f g$$

depends on V_o



$$V_f = \frac{F_b}{\rho_f g}$$

V_o

مسألة 1.7

1.7

$9.5 \times 10^2 \text{ N}$ metal in air and $7 \times 10^2 \text{ N}$ immersed in fluid of $\rho_f = 1000$.

The density of object

$$V_o = \frac{F_b}{\rho_f g} = \frac{2.5 \times 10^2}{1000 \times 9.8}$$

$$F_b = \frac{w_o \rho_f}{\rho_o}$$

$$\rho_o = \frac{w_o}{F_b} \rho_f = \frac{9.5 \times 10^2}{2.5 \times 10^2} \times 1000 = 3800$$

The volume of object

$$V_o = \frac{w_o}{\rho_o g} = \frac{9.5 \times 10^2}{3800 \times 9.8}$$